

#####

STARTING CONDITION C TEST RUN

Total problems: 15

#####

#####

Running problem 1/15

#####

=====
=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-
conditioned solve

Problem ID: P01

Problem: Add $1/2 + 1/5$

=====
=====

===== Retrieved Method Context =====

Score: 0.69088745 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-decimals__m_mixed_number-p002

Score: 0.66497064 | Topic: f_fractions_addition | Domain: fractions-decimals | Doc: fractions-decimals__f_fractions_addition-p002

Score: 0.6509206 | Topic: f_fractions_addition | Domain: fractions-decimals | Doc: fractions-decimals__f_fractions_addition-p001

=====

===== Retrieval Gate =====

Top retrieval score: 0.69088745

Threshold: 0.67

Use retrieval: True

Retreival score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['fractions-decimals__f_fractions_addition-p002', 'fractions-decimals__f_fractions_addition-p001']

Method guidance: To add fractions with different denominators, first find the least common denominator (LCD). Rewrite each fraction with this common denominator by multiplying both the numerator and denominator. Then, add the numerators and keep the common denominator. Finally, simplify the resulting fraction if possible.

LaTeX detail instruction: Show the steps for finding the LCD, rewriting fractions with the common denominator, adding the numerators, and simplifying the final fraction.

=====

$$\frac{1}{2} + \frac{1}{5}$$

To add fractions, find the least common denominator (LCD).

The denominators are 2 and 5. The LCD of 2 and 5 is 10.

$$\frac{1}{2} = \frac{1 \times 5}{2 \times 5} = \frac{5}{10}$$

$$\frac{1}{5} = \frac{1 \times 2}{5 \times 2} = \frac{2}{10}$$

Now, add the fractions with the common denominator:

$$\frac{5}{10} + \frac{2}{10} = \frac{5 + 2}{10}$$

$$= \frac{7}{10}$$

#####

Running problem 2/15

#####

=====
=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-conditioned solve

Problem ID: P02

Problem: Multiply $\frac{3}{4}$ by $\frac{2}{5}$

=====

===== Retrieved Method Context =====

Score: 0.70102024 | Topic: e_equivalent_fractions | Domain: fractions-decimals | Doc: fractions-decimals__e_equivalent_fractions-p002

Score: 0.67506814 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-decimals__m_mixed_number-p002

Score: 0.66949725 | Topic: f_fractions_multiplication | Domain: fractions-decimals | Doc: fractions-decimals__f_fractions_multiplication-p002

=====

===== Retrieval Gate =====

Top retrieval score: 0.70102024

Threshold: 0.67

Use retrieval: True

Retreival score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['fractions-decimals__f_fractions_multiplication-p002']

Method guidance: To multiply fractions, multiply the numerators together and multiply the denominators together. Then, simplify the resulting fraction if possible.

LaTeX detail instruction: Show the multiplication of numerators and denominators, then the simplification step.

=====

$$\begin{aligned}\frac{3}{4} \times \frac{2}{5} &= \frac{3 \times 2}{4 \times 5} && \text{Multiply numerators and denominators} \\ &= \frac{6}{20} \\ &= \frac{6 \div 2}{20 \div 2} && \text{Simplify by dividing by the common factor of 2} \\ &= \frac{3}{10}\end{aligned}$$

#####

Running problem 3/15

#####

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-conditioned solve

Problem ID: P03

Problem: The fraction 3/4 written as a decimal is

=====

===== Retrieved Method Context =====

Score: 0.7162106 | Topic: d_decimal_fraction_percentage | Domain: fractions-decimals |
Doc: fractions-decimals__d_decimal_fraction_percentage-p001

Score: 0.71258044 | Topic: c_converting_fractions_decimals | Domain: fractions-decimals |
Doc: fractions-decimals__c_converting_fractions_decimals-p001

Score: 0.7060766 | Topic: d_decimal | Domain: fractions-decimals | Doc: fractions-
decimals__d_decimal-p001

=====

===== Retrieval Gate =====

Top retrieval score: 0.7162106

Threshold: 0.67

Use retrieval: True

Retreival score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['fractions-decimals__d_decimal_fraction_percentage-p001', 'fractions-
decimals__c_converting_fractions_decimals-p001']

Method guidance: To convert a fraction to a decimal, divide the numerator by the denominator. This process results in either a terminating or repeating decimal.

LaTeX detail instruction: Show the division of the numerator by the denominator to convert the fraction to a decimal.

=====

$$\frac{3}{4} = 3 \div 4$$
$$= 0.75$$

#####

Running problem 4/15

#####

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-conditioned solve

Problem ID: P04

Problem: If $R = \{a,b,c,d,e,f\}$, draw the set that is not a subset of R

=====

===== Retrieved Method Context =====

Score: 0.67028904 | Topic: s_subset | Domain: set-theory | Doc: set-theory__s_subset-p002

Score: 0.668005 | Topic: i_intersection | Domain: set-theory | Doc: set-theory__i_intersection-p002

Score: 0.6619897 | Topic: p_proper_subset | Domain: set-theory | Doc: set-theory__p_proper_subset-p001

=====

===== Retrieval Gate =====

Top retrieval score: 0.67028904

Threshold: 0.67

Use retrieval: True

Retrieval score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['set-theory__s_subset-p002']

Method guidance: To determine if a set is not a subset of R , you need to find at least one element in that set that is not present in R . If every element of a set is also an element of R , then it is a subset of R .

LaTeX detail instruction: Explain the definition of a subset and how to identify a set that is not a subset by finding a counterexample.

=====

$$\frac{3}{4} = 3 \div 4$$

$$= 0.75$$

#####

#####

Running problem 5/15

#####

#####

=====

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-
conditioned solve

Problem ID: P05

Problem: Draw a rectangular shape that shows the shading of 1/3

=====

=====

===== Retrieved Method Context =====

Score: 0.6185677 | Topic: f_fraction_number_line | Domain: fractions-decimals | Doc:
fractions-decimals__f_fraction_number_line-p001

Score: 0.61370826 | Topic: d_decimal_fraction_percentage | Domain: fractions-decimals |
Doc: fractions-decimals__d_decimal_fraction_percentage-p003

Score: 0.6094401 | Topic: c_converting_fractions_decimals | Domain: fractions-decimals |
Doc: fractions-decimals__c_converting_fractions_decimals-p002

=====

===== Direct Visual Renderer =====

Detected visual kind: fraction_bar

Skipped guidance selector and solver LLM for this visual-only prompt.

Rendering with matplotlib.

=====

Fraction Bar / Area Model



1/3 shaded

#####

Running problem 6/15

#####

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-
conditioned solve

Problem ID: P06

Problem: Show that $3 * 7 + 3 * 9$ is the same as $7 + 7 + 7 + 9 + 9 + 9$

=====

===== Retrieved Method Context =====

Score: 0.62641335 | Topic: e_exponentiation | Domain: arithmetic-operations | Doc: arithmetic-operations__e_exponentiation-p001

Score: 0.6248069 | Topic: p_product | Domain: arithmetic-operations | Doc: arithmetic-operations__p_product-p001

Score: 0.62218285 | Topic: s_sum | Domain: arithmetic-operations | Doc: arithmetic-operations__s_sum-p001

=====

===== Retrieval Gate =====

Top retrieval score: 0.62641335

Threshold: 0.67

Use retrieval: False

Skipped guidance selector LLM call.

=====

===== Selected Retrieval Guidance =====

Use retrieval: False

Selected docs: []

Method guidance:

LaTeX detail instruction:

=====

$$\begin{aligned} 3 \times 7 + 3 \times 9 &= (7 + 7 + 7) + (9 + 9 + 9) \\ &= 21 + 27 \\ &= 48 \end{aligned}$$

#####

Running problem 7/15

#####

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-conditioned solve

Problem ID: P07

Problem: In a proper fraction, the difference between the numerator and denominator is 5. If the denominator is 7, what is the fraction?

=====

===== Retrieved Method Context =====

Score: 0.77135897 | Topic: p_proper_fraction | Domain: fractions-decimals | Doc: fractions-decimals__p_proper_fraction-p002

Score: 0.76488376 | Topic: p_proper_fraction | Domain: fractions-decimals | Doc: fractions-decimals__p_proper_fraction-p001

Score: 0.71748185 | Topic: p_proper_fraction | Domain: fractions-decimals | Doc: fractions-decimals__p_proper_fraction-p003

=====

===== Retrieval Gate =====

Top retrieval score: 0.77135897

Threshold: 0.67

Use retrieval: True

Retrieval score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['fractions-decimals__p_proper_fraction-p001', 'fractions-decimals__p_proper_fraction-p002', 'fractions-decimals__p_proper_fraction-p003']

Method guidance: To solve this problem, first define a proper fraction. A proper fraction has a numerator that is smaller than its denominator. Use the given information about the denominator and the difference between the numerator and denominator to find the numerator. Then, form the fraction.

LaTeX detail instruction: Define a proper fraction. Calculate the numerator using the given difference and denominator. Form the fraction.

=====

A proper fraction has a numerator smaller than its denominator.

Given:

Denominator = 7

Difference between denominator and numerator = 5

Let the numerator be n .

Denominator – Numerator = 5

$$7 - n = 5$$

$$n = 7 - 5$$

$$n = 2$$

The numerator is 2 and the denominator is 7.

The fraction is $\frac{2}{7}$.

Since $2 < 7$, it is a proper fraction.

#####

Running problem 8/15

#####

=====
=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-
conditioned solve

Problem ID: P08

Problem: What is 1/8 of 16

=====

===== Retrieved Method Context =====

Score: 0.59877014 | Topic: c_converting_fractions_decimals | Domain: fractions-decimals |
Doc: fractions-decimals__c_converting_fractions_decimals-p002

Score: 0.59734344 | Topic: t_thousandth | Domain: fractions-decimals | Doc: fractions-
decimals__t_thousandth-p001

Score: 0.5958979 | Topic: f_fraction_decimal_chart | Domain: fractions-decimals | Doc:
fractions-decimals__f_fraction_decimal_chart-p002

=====

===== Retrieval Gate =====

Top retrieval score: 0.59877014

Threshold: 0.67

Use retrieval: False

Skipped guidance selector LLM call.

=====

===== Selected Retrieval Guidance =====

Use retrieval: False

Selected docs: []

Method guidance:

LaTeX detail instruction:

=====

$$\begin{aligned}\frac{1}{8} \text{ of } 16 &= \frac{1}{8} \times 16 \\ &= \frac{16}{8} \\ &= 2\end{aligned}$$

#####

Running problem 9/15

#####

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-
conditioned solve

Problem ID: P09

Problem: The sum of 0.5 and 1.7 is

=====

===== Retrieved Method Context =====

Score: 0.6196549 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-
decimals__m_mixed_number-p002

Score: 0.61845994 | Topic: s_sum | Domain: arithmetic-operations | Doc: arithmetic-operations__s_sum-p001

Score: 0.6165011 | Topic: i_improper_fraction | Domain: fractions-decimals | Doc: fractions-decimals__i_improper_fraction-p001

=====

===== Retrieval Gate =====

Top retrieval score: 0.6196549

Threshold: 0.67

Use retrieval: False

Skipped guidance selector LLM call.

=====

===== Selected Retrieval Guidance =====

Use retrieval: False

Selected docs: []

Method guidance:

LaTeX detail instruction:

=====

$0.5 + 1.7 = 2.2$

#####

Running problem 10/15

#####

=====
=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-
conditioned solve

Problem ID: P10

Problem: The top length is a square is 7cm, what is the perimeter?

=====
=====

===== Retrieved Method Context =====

Score: 0.6843953 | Topic: p_perimeter | Domain: area-perimeter | Doc: area-
perimeter__p_perimeter-p002

Score: 0.66611886 | Topic: m_mensuration | Domain: area-perimeter | Doc: area-
perimeter__m_mensuration-p001

Score: 0.6581342 | Topic: p_perimeter | Domain: area-perimeter | Doc: area-
perimeter__p_perimeter-p001

=====

===== Retrieval Gate =====

Top retrieval score: 0.6843953

Threshold: 0.67

Use retrieval: True

Retreival score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['area-perimeter__p_perimeter-p001', 'area-perimeter__p_perimeter-p002']

Method guidance: To find the perimeter of a square, identify that all four sides are equal in length. Then, sum the lengths of all four sides.

LaTeX detail instruction: Define perimeter as the sum of side lengths. State that a square has four equal sides. Show the calculation of the sum of the four side lengths.

=====

The perimeter of a square is the sum of the lengths of its four sides.

Since all sides of a square are equal, and one side is 7 cm,
each of the four sides is 7 cm long.

$$\begin{aligned}\text{Perimeter} &= \text{side} + \text{side} + \text{side} + \text{side} \\ &= 7 \text{ cm} + 7 \text{ cm} + 7 \text{ cm} + 7 \text{ cm} \\ &= 4 \times 7 \text{ cm} \\ &= 28 \text{ cm}\end{aligned}$$

#####

Running problem 11/15

#####

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-conditioned solve

Problem ID: P11

Problem: Show the largest to the smallest fraction and why they are so, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{5}$

=====

===== Retrieved Method Context =====

Score: 0.70217276 | Topic: e_egyptian_fraction | Domain: fractions-decimals | Doc: fractions-decimals__e_egyptian_fraction-p001

Score: 0.69676995 | Topic: u_unit_fraction | Domain: fractions-decimals | Doc: fractions-decimals__u_unit_fraction-p002

Score: 0.6870134 | Topic: s_simplifying_fractions | Domain: fractions-decimals | Doc: fractions-decimals__s_simplifying_fractions-p002

=====

===== Retrieval Gate =====

Top retrieval score: 0.70217276

Threshold: 0.67

Use retrieval: True

Retreival score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['fractions-decimals__u_unit_fraction-p002']

Method guidance: To compare unit fractions, understand that a larger denominator means a smaller unit fraction because the whole is split into more pieces.

LaTeX detail instruction: Explain that for unit fractions, the fraction with the largest denominator is the smallest, and the fraction with the smallest denominator is the largest.

=====

To compare unit fractions, the fraction with the largest denominator is the smallest, and the fraction with the smallest denominator is the largest.

Given fractions: $\frac{1}{2}, \frac{1}{4}, \frac{1}{3}, \frac{1}{5}$

Comparing the denominators: 2, 4, 3, 5.

Ordering the denominators from smallest to largest: $2 < 3 < 4 < 5$.

Therefore, ordering the fractions from largest to smallest:

$$\frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \frac{1}{5}$$

#####

Running problem 12/15

#####

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-conditioned solve

Problem ID: P12

Problem: Draw $A \cup B$ if $A = \{1,5,6\}$, $B = \{a,c,d\}$

=====

===== Retrieved Method Context =====

Score: 0.6957648 | Topic: c_cup | Domain: set-theory | Doc: set-theory__c_cup-p002

Score: 0.6882458 | Topic: u_union | Domain: set-theory | Doc: set-theory__u_union-p001

Score: 0.6838386 | Topic: u_union | Domain: set-theory | Doc: set-theory__u_union-p002

=====

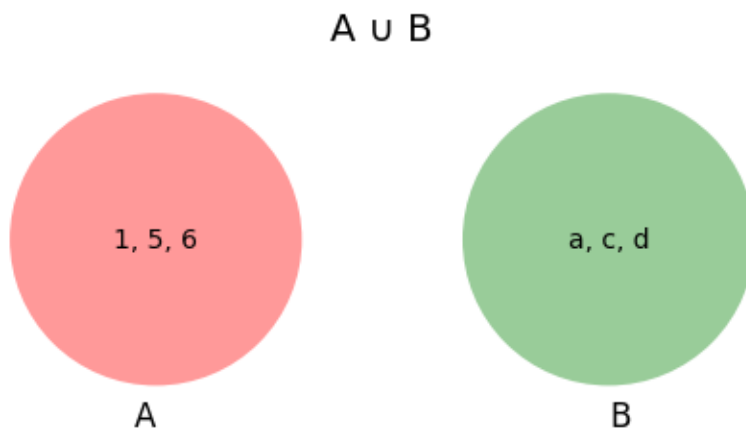
===== Direct Visual Renderer =====

Detected visual kind: venn2

Skipped guidance selector and solver LLM for this visual-only prompt.

Rendering with matplotlib.

=====



#####

Running problem 13/15

#####

=====
=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-
conditioned solve

Problem ID: P13

Problem: Set A = {Even prime numbers greater than 2}, how many members does set A
have?

=====
=====

===== Retrieved Method Context =====

Score: 0.6831758 | Topic: s_set_braces | Domain: set-theory | Doc: set-
theory__s_set_braces-p001

Score: 0.67578673 | Topic: c_countable | Domain: set-theory | Doc: set-
theory__c_countable-p001

Score: 0.6723573 | Topic: e_element_of_a_set | Domain: set-theory | Doc: set-
theory__e_element_of_a_set-p001

=====

===== Retrieval Gate =====

Top retrieval score: 0.6831758

Threshold: 0.67

Use retrieval: True

Retrieval score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['set-theory__s_set_braces-p001', 'set-theory__e_element_of_a_set-p001']

Method guidance: To solve this problem, first identify the elements that satisfy the given conditions for set A. The problem asks for even prime numbers greater than 2. Recall the definition of a prime number (a natural number greater than 1 that has no positive divisors other than 1 and itself) and an even number (an integer divisible by 2). Once the elements are identified, list them within set braces. Finally, count the number of elements in the set to determine its cardinality.

LaTeX detail instruction: List the elements of set A and then state the cardinality of the set.

=====

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

The first few prime numbers are 2, 3, 5, 7, 11, ...

An even number is an integer divisible by 2.

We are looking for even prime numbers greater than 2.

The only even prime number is 2.

Since we need prime numbers greater than 2, there are no even prime numbers greater than 2.

$A = \text{Even prime numbers greater than 2}$

$A = \emptyset$

The number of members in set A is the cardinality of A, denoted as $|A|$.

$|A| = 0$

#####

Running problem 14/15

#####

=====
=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-
conditioned solve

Problem ID: P14

Problem: Draw a rectangle having 6 regions inside, with 1/2 shaded

=====
=====

===== Retrieved Method Context =====

Score: 0.6007097 | Topic: q_quotient | Domain: arithmetic-operations | Doc: arithmetic-
operations__q_quotient-p003

Score: 0.6007097 | Topic: s_sum | Domain: arithmetic-operations | Doc: arithmetic-
operations__s_sum-p003

Score: 0.60047984 | Topic: a_area_irregular_polygons | Domain: area-perimeter | Doc:
area-perimeter__a_area_irregular_polygons-p001

=====

===== Direct Visual Renderer =====

Detected visual kind: box_grid

Skipped guidance selector and solver LLM for this visual-only prompt.

Rendering with matplotlib.

=====

Rectangle divided into 6 equal regions



$\frac{1}{2}$ shaded

#####

Running problem 15/15

#####

=====

CONDITION C: Full Pipeline: Classification/retrieval, guidance selection, method-conditioned solve

Problem ID: P15

Problem: The mixed number $7\frac{2}{5}$ written as an improper fraction is

=====

===== Retrieved Method Context =====

Score: 0.7374382 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-decimals__m_mixed_number-p003

Score: 0.7227895 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-decimals__m_mixed_number-p001

Score: 0.72033834 | Topic: i_improper_fraction | Domain: fractions-decimals | Doc: fractions-decimals__i_improper_fraction-p001

=====

===== Retrieval Gate =====

Top retrieval score: 0.7374382

Threshold: 0.67

Use retrieval: True

Retrieval score satisfied.

=====

===== Selected Retrieval Guidance =====

Use retrieval: True

Selected docs: ['fractions-decimals__m_mixed_number-p003', 'fractions-decimals__m_mixed_number-p001']

Method guidance: To convert a mixed number to an improper fraction, multiply the whole number by the denominator, then add the numerator. Place this result over the original denominator.

LaTeX detail instruction: Show the steps for converting a mixed number to an improper fraction: multiply the whole number by the denominator, add the numerator, and place the result over the original denominator.

=====

$$\begin{aligned}
 7\frac{2}{5} &= \frac{(7 \times 5) + 2}{5} && \text{Multiply the whole number by the denominator, then add the numerator.} \\
 &= \frac{35 + 2}{5} \\
 &= \frac{37}{5} && \text{Place the result over the original denominator.}
 \end{aligned}$$

#####

FINISHED CONDITION C TEST RUN

#####